

| May 2016 |    |   |    |    |    |  | June 2016 |   |    |    |    |   |   | July 2016 |    |    |    |  |  |  |
|----------|----|---|----|----|----|--|-----------|---|----|----|----|---|---|-----------|----|----|----|--|--|--|
| M        | 30 | 2 | 9  | 16 | 23 |  | M         | 6 | 13 | 20 | 27 | M | 4 | 11        | 18 | 25 |    |  |  |  |
| T        | 31 | 3 | 10 | 17 | 24 |  | T         | 7 | 14 | 21 | 28 | T | 5 | 12        | 19 | 26 |    |  |  |  |
| W        |    | 4 | 11 | 18 | 25 |  | W         | 1 | 8  | 15 | 22 | W | 6 | 13        | 20 | 27 |    |  |  |  |
| T        |    | 5 | 12 | 19 | 26 |  | T         | 2 | 9  | 16 | 23 | T | 7 | 14        | 21 | 28 |    |  |  |  |
| F        |    | 6 | 13 | 20 | 27 |  | F         | 3 | 10 | 17 | 24 | F | 1 | 8         | 15 | 22 |    |  |  |  |
| S        |    | 7 | 14 | 21 | 28 |  | S         | 4 | 11 | 18 | 25 | S | 2 | 9         | 16 | 23 |    |  |  |  |
| S        | 1  | 8 | 15 | 22 | 29 |  | S         | 5 | 12 | 19 | 26 | S | 3 | 10        | 17 | 24 | 31 |  |  |  |

# Worksheet #06

April

WEEK 14

Adil Abbas  
BSAI Sec D,  
5000001135.

097/269 Wednesday 6

## Calculus & Analytical Geometry:

8.00 am

⇒ Evaluate the integral by making the given substitution.

8.30

1,  $\int \cos 2x \, dx$ ,  $u = 2x$

9.00

Solution ::

9.30

$$\frac{du}{dx} = 2$$

10.00

$$du = 2 \, dx$$

10.30

$$\boxed{\frac{du}{2} = dx}$$

11.00

11.30

$$\int \cos(u) \frac{du}{2}$$

Noon

$$\frac{1}{2} \int \cos(u) \, du$$

12.30

$$\frac{1}{2} \sin(u) + C$$

1.00

$$= \boxed{\frac{1}{2} \sin 2x + C} \text{ Ans.}$$

1.30

4,  $\int \sin^2 \theta \cos \theta \, d\theta$ ,  $u = \sin \theta$

2.30

Solution ::

$$\frac{du}{d\theta} = \cos \theta$$

3.00

$$\boxed{du = \cos \theta \, d\theta}$$

3.30

$$\int (u)^2 \, du$$

4.00

$$\frac{(u)^3}{3} + C$$

4.30

$$\frac{(\sin \theta)^3}{3} + C$$

$$\boxed{\frac{\sin^3 \theta}{3} + C} \text{ Ans.}$$

5.00

6.00 pm

April  
WEEK 14

February 2016

|   |   |    |    |    |    |
|---|---|----|----|----|----|
| M | 1 | 8  | 15 | 22 | 29 |
| T | 2 | 9  | 16 | 23 |    |
| W | 3 | 10 | 17 | 24 |    |
| T | 4 | 11 | 18 | 25 |    |
| F | 5 | 12 | 19 | 26 |    |
| S | 6 | 13 | 20 | 27 |    |
| S | 7 | 14 | 21 | 28 |    |

March 2016

|   |   |    |    |    |
|---|---|----|----|----|
| M | 7 | 14 | 21 | 28 |
| T | 1 | 8  | 15 | 22 |
| W | 2 | 9  | 16 | 23 |
| T | 3 | 10 | 17 | 24 |
| F | 4 | 11 | 18 | 25 |
| S | 5 | 12 | 19 | 26 |
| S | 6 | 13 | 20 | 27 |

April 2016

|   |   |    |    |    |
|---|---|----|----|----|
| M | 4 | 11 | 18 | 25 |
| T | 5 | 12 | 19 | 26 |
| W | 6 | 13 | 20 | 27 |
| T | 7 | 14 | 21 | 28 |
| F | 1 | 8  | 15 | 22 |
| S | 2 | 9  | 16 | 23 |
| S | 3 | 10 | 17 | 24 |

7 Thursday 098/268

8.00 am

3)  $\int x^2 \sqrt{x^3+1} dx, u = x^3+1$

8.30

Solution:

9.00  $\frac{du}{dx} = 3x^2+0$

9.30  $du = 3x^2 dx$

10.00  $\frac{du}{3} = x^2 dx$

10.30

$\int x^2 \sqrt{x^3+1} dx$

11.00

$\int \sqrt{x^3+1} x^2 dx$

11.30

$\int \sqrt{u} \frac{du}{3}$

Noon

$\frac{1}{3} \int u^{1/2} du$

12.30

$\frac{1}{3} \left[ \frac{u^{1/2+1}}{1/2+1} \right] + C$

1.00

$\frac{1}{3} \left[ \frac{u^{3/2}}{3/2} \right] + C$

1.30

$\frac{1}{3} \cdot \frac{2}{3} u^{3/2} + C$

2.00

$\frac{2}{9} (u)^{3/2} + C$

2.30

$\frac{2}{9} (x^3+1)^{3/2} + C$

3.00

$\frac{2}{9} (x^3+1)^{3/2} + C$

3.30

Ans //

4.00

4.30

5.00

6.00 pm

5)  $\int \frac{x^3}{x^4-5} dx, u = x^4-5$

Solution:

$\frac{du}{dx} = 4x^3$

$du = 4x^3 dx$

$\frac{du}{4} = x^3 dx$

$\frac{1}{4} \int \frac{dx}{u}$

$\frac{1}{4} \ln |u| + C$

$\frac{1}{4} \ln |x^4-5| + C$

Ans //

| May 2016 |    |   |    |    | June 2016 |   |   |    |    | July 2016 |   |   |    |    |    |
|----------|----|---|----|----|-----------|---|---|----|----|-----------|---|---|----|----|----|
| M        | 30 | 2 | 9  | 16 | 23        | M | 6 | 13 | 20 | 27        | M | 4 | 11 | 18 | 25 |
| T        | 31 | 3 | 10 | 17 | 24        | T | 7 | 14 | 21 | 28        | T | 5 | 12 | 19 | 26 |
| W        |    | 4 | 11 | 18 | 25        | W | 1 | 8  | 15 | 22        | W | 6 | 13 | 20 | 27 |
| T        |    | 5 | 12 | 19 | 26        | T | 2 | 9  | 16 | 23        | T | 7 | 14 | 21 | 28 |
| F        |    | 6 | 13 | 20 | 27        | F | 3 | 10 | 17 | 24        | F | 1 | 8  | 15 | 22 |
| S        |    | 7 | 14 | 21 | 28        | S | 4 | 11 | 18 | 25        | S | 2 | 9  | 16 | 23 |
| S        | 1  | 8 | 15 | 22 | 29        | S | 5 | 12 | 19 | 26        | S | 3 | 10 | 17 | 24 |

099/267 Friday 8

→ Evaluate the indefinite integral.

8.30 9)  $\int x \sqrt{1-x^2} dx$

9.00 Solution:-

=  $u = 1-x^2$

8.30  $\frac{du}{dx} = -2x$

10.00  $du = -2x dx$

=  $\frac{du}{-2} = x dx$

11.00  $\int x \sqrt{1-x^2} dx$

11.30  $\int \sqrt{1-x^2} x dx$

Noon  $-\frac{1}{2} \int \sqrt{u} du$

12.30  $-\frac{1}{2} \frac{u^{1/2+1}}{1/2+1} + C$

1.00  $-\frac{1}{2} \frac{u^{3/2}}{3/2} + C$

2.00  $-\frac{1}{2} \times \frac{2}{3} (u)^{3/2} + C$

2.30  $-\frac{1}{3} (u)^{3/2} + C$

3.00  $-\frac{1}{3} (1-x^2)^{3/2} + C$

3.30 Ans

4.00

4.30

5.00

6.00 pm

10)  $\int (5-3x)^{10} dx$

Solution:-

$u = 5-3x$

$\frac{du}{dx} = -3$

$du = -3 dx$

$\frac{du}{-3} = dx$

$\int (5-3x)^{10} dx$

$-\frac{1}{3} \int (u)^{10} dx$

$-\frac{1}{3} \frac{(u)^{10+1}}{10+1} + C$

$-\frac{1}{3} \frac{(u)^{11}}{11} + C$

$-\frac{1}{33} (u)^{11} + C$

$-\frac{1}{33} (5-3x)^{11} + C$

Ans

|   |   |    |    |    |    |
|---|---|----|----|----|----|
| M | 1 | 8  | 15 | 22 | 29 |
| T | 2 | 9  | 16 | 23 |    |
| W | 3 | 10 | 17 | 24 |    |
| T | 4 | 11 | 18 | 25 |    |
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| S | 6 | 13 | 20 | 27 |    |
| S | 7 | 14 | 21 | 28 |    |

|   |   |    |    |    |    |
|---|---|----|----|----|----|
| M |   | 7  | 14 | 21 | 28 |
| T | 1 | 8  | 15 | 22 | 29 |
| W | 2 | 9  | 16 | 23 | 30 |
| T | 3 | 10 | 17 | 24 | 31 |
| F | 4 | 11 | 18 | 25 |    |
| S | 5 | 12 | 19 | 26 |    |
| S | 6 | 13 | 20 | 27 |    |

|   |   |    |    |    |    |
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| T |   | 5  | 12 | 19 | 26 |
| W |   | 6  | 13 | 20 | 27 |
| T |   | 7  | 14 | 21 | 28 |
| F | 1 | 8  | 15 | 22 | 29 |
| S | 2 | 9  | 16 | 23 | 30 |
| S | 3 | 10 | 17 | 24 |    |

9 Saturday 100/266

Day of Valor (Philippines)

8.00 am

$$11, \int t^3 e^{-t^4} dt$$

8.30

Solution r

$$9.00 \quad u = -t^4$$

$$9.30 \quad \frac{du}{dt} = -4t^3$$

$$10.00 \quad dx = -4t^3 dt$$

$$10.30 \quad \frac{dx}{-4} = t^3 dt$$

$$11.00 \quad \int t^3 e^{-t^4} dt$$

11.30

$$\int e^{-t^4} t^3 dt$$

Noon

$$-1/4 \int e^u du$$

12.30

$$1.00 \quad -\frac{1}{4} e^u + c$$

1.30

$$-\frac{1}{4} e^{-t^4} + c$$

2.00

2.30

Ans,,

3.00

3.30

4.00

4.30

5.00

6.00 pm

$$12, \int \sin t \sqrt{1 + \cos t} dt$$

Solution r

$$u = 1 + \cos t$$

$$\frac{du}{dt} = -\sin t$$

By Multiplying -1 on b.s,,

$$-1(du) = -1(-\sin t dt)$$

$$-du = \sin t dt$$

$$\int \sqrt{u} (-du)$$

$$-\int \sqrt{u} du$$

$$- \frac{(u)^{1/2+1}}{1/2+1} + c$$

$$- \frac{u^{3/2}}{3/2} + c$$

$$- \frac{2}{3} (u)^{3/2} + c$$

$$-\frac{2}{3} (1 + \cos t)^{3/2} + c$$

Ans,,



| May 2016 |    |   |    |    |    |  | June 2016 |   |    |    |    |    |  | July 2016 |   |    |    |    |    |  |
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| M        | 30 | 2 | 9  | 16 | 23 |  | M         | 6 | 13 | 20 | 27 |    |  | M         | 4 | 11 | 18 | 25 |    |  |
| T        | 31 | 3 | 10 | 17 | 24 |  | T         | 7 | 14 | 21 | 28 |    |  | T         | 5 | 12 | 19 | 26 |    |  |
| W        |    | 4 | 11 | 18 | 25 |  | W         | 1 | 8  | 15 | 22 | 29 |  | W         | 6 | 13 | 20 | 27 |    |  |
| T        |    | 5 | 12 | 19 | 26 |  | T         | 2 | 9  | 16 | 23 | 30 |  | T         | 7 | 14 | 21 | 28 |    |  |
| F        |    | 6 | 13 | 20 | 27 |  | F         | 3 | 10 | 17 | 24 |    |  | F         | 1 | 8  | 15 | 22 | 29 |  |
| S        |    | 7 | 14 | 21 | 28 |  | S         | 4 | 11 | 18 | 25 |    |  | S         | 2 | 9  | 16 | 23 | 30 |  |
| S        | 1  | 8 | 15 | 22 | 29 |  | S         | 5 | 12 | 19 | 26 |    |  | S         | 3 | 10 | 17 | 24 | 31 |  |

April  
WEEK 14

101/265 Sunday 10

8.00 am

$$13, \int \sin\left(\frac{\pi t}{3}\right) dt$$

8.30

Solution:

9.00

$$u = \frac{\pi t}{3}$$

9.30

$$\frac{du}{dt} = \frac{\pi}{3}$$

10.00

$$du = \frac{\pi}{3} dt$$

10.30

$$\frac{3 du}{\pi} = dt$$

11.30

Noon

$$\int \sin\left(\frac{\pi t}{3}\right) dt$$

12.30

$$\int \sin(u) \frac{3}{\pi} du$$

1.00

$$-\frac{3}{\pi} \int \sin(u) du$$

1.30

$$-\frac{3}{\pi} (-\cos(u)) + C$$

2.00

2.30

$$-\frac{3}{\pi} \cos(u) + C$$

3.00

3.30

$$-\frac{3}{\pi} \cos\left(\frac{\pi t}{3}\right) + C$$

4.00

4.30

5.00

6.00 pm

$$14, \int \sec^2 2\theta d\theta$$

Solution:

$$u = 2\theta$$

$$\frac{du}{d\theta} = 2$$

$$du = 2 d\theta$$

$$\frac{du}{2} = d\theta$$

$$\int \sec^2 2\theta d\theta$$

$$\frac{1}{2} \int \sec^2(u) du$$

$$\frac{1}{2} \tan(u) + C$$

$$\boxed{\frac{1}{2} \tan(2\theta) + C}$$

Ans

|   |   |    |    |    |    |
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| T | 4 | 11 | 18 | 25 |    |
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| S | 7 | 14 | 21 | 28 |    |

|   |    |    |    |    |
|---|----|----|----|----|
| M | 7  | 14 | 21 | 28 |
| T | 8  | 15 | 22 | 29 |
| W | 9  | 16 | 23 | 30 |
| T | 10 | 17 | 24 | 31 |
| F | 11 | 18 | 25 |    |
| S | 12 | 19 | 26 |    |
| S | 13 | 20 | 27 |    |

|   |   |    |    |    |
|---|---|----|----|----|
| M | 4 | 11 | 18 | 25 |
| T | 5 | 12 | 19 | 26 |
| W | 6 | 13 | 20 | 27 |
| T | 7 | 14 | 21 | 28 |
| F | 1 | 8  | 15 | 22 |
| S | 2 | 9  | 16 | 23 |
| S | 3 | 10 | 17 | 24 |

11 Monday 102/264

8.00 am

15)  $\int \frac{dx}{4x+7}$

8.30

Solution:

9.00

$$u = 4x + 7$$

9.30

$$\frac{du}{dx} = 4$$

10.00

$$du = 4 dx$$

10.30

$$\frac{du}{4} = dx$$

11.00

$$\int \frac{dx}{4x+7}$$

Noon

$$\int \frac{1}{u} \cdot \frac{du}{4}$$

12.30

$$\frac{1}{4} \int \frac{1}{u} \cdot du$$

1.30

$$= \frac{1}{4} \ln |u| + c$$

2.00

$$\boxed{\frac{1}{4} \ln |4x+7| + c}$$

3.00

Ans //

3.30

4.00

4.30

5.00

6.00 pm

16)  $\int y^2 (4-y^3)^{2/3} dy$

Solution:

$$u = 4 - y^3$$

$$\frac{du}{dy} = -3y^2$$

$$du = -3y^2 dy$$

$$\frac{du}{-3} = y^2 dy$$

$$\int y^2 (4-y^3)^{2/3} dy$$

$$-\frac{1}{3} \int (u)^{2/3} du$$

$$= -\frac{1}{3} \frac{(u)^{2/3+1}}{2/3+1} + c$$

$$= -\frac{1}{3} \frac{(u)^{5/3}}{5/3} + c$$

$$= -\frac{1}{3} \cdot \frac{3}{5} (u)^{5/3} + c$$

$$= -\frac{1}{5} (u)^{5/3} + c$$

$$= \boxed{-\frac{1}{5} (4-y^3)^{5/3} + c}$$

Ans //

May 2016

|   |    |   |    |    |    |   |
|---|----|---|----|----|----|---|
| M | 30 | 2 | 9  | 16 | 23 | M |
| T | 31 | 3 | 10 | 17 | 24 | T |
| W |    | 4 | 11 | 18 | 25 | W |
| T |    | 5 | 12 | 19 | 26 | T |
| F |    | 6 | 13 | 20 | 27 | F |
| S |    | 7 | 14 | 21 | 28 | S |
| S | 1  | 8 | 15 | 22 | 29 | S |

June 2016

|   |    |    |    |    |
|---|----|----|----|----|
|   | 6  | 13 | 20 | 27 |
|   | 7  | 14 | 21 | 28 |
| 1 | 8  | 15 | 22 | 29 |
| 2 | 9  | 16 | 23 | 30 |
| 3 | 10 | 17 | 24 |    |
| 4 | 11 | 18 | 25 |    |
| 5 | 12 | 19 | 26 |    |

July 2016

|   |   |    |    |    |
|---|---|----|----|----|
| M | 4 | 11 | 18 | 25 |
| T | 5 | 12 | 19 | 26 |
| W | 6 | 13 | 20 | 27 |
| T | 7 | 14 | 21 | 28 |
| F | 1 | 8  | 15 | 22 |
| S | 2 | 9  | 16 | 23 |
| S | 3 | 10 | 17 | 24 |

April  
WEEK 15

103/263 Tuesday 12

8.00 am

$$17, \int \frac{\cos \theta}{1 + \sin \theta} d\theta$$

8.30

Solution :-

9.00

$$u = 1 + \sin \theta$$

9.30

$$\frac{du}{d\theta} = \cos \theta$$

10.00

$$du = \cos \theta d\theta$$

10.30

$$\int \frac{\cos \theta}{1 + \sin \theta} d\theta$$

11.00

$$\int \frac{du}{u}$$

11.30

Noon

$$= \ln |u| + c$$

12.30

$$= \boxed{\ln |1 + \sin \theta| + c}$$

1.00

Ans 4

1.30

2.00

2.30

3.00

3.30

4.00

4.30

5.00

6.00 pm

$$18, \int \frac{z^2}{z^3 + 1} dz$$

Solution :-

$$u = z^3 + 1$$

$$\frac{du}{dz} = 3z^2$$

$$du = 3z^2 dz$$

$$\frac{du}{3} = z^2 dz$$

$$\int \frac{z^2}{z^3 + 1} dz$$

$$\frac{1}{3} \int \frac{1}{u} \cdot du$$

$$= \frac{1}{3} \cdot \ln |u| + c$$

$$= \boxed{\frac{1}{3} \cdot \ln |z^3 + 1| + c}$$

Ans 4